

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech. (CSE) Semester – I Examination, Jan 2010-11
EC – 701 Advanced High speed Network

Date: 03/ 01/11, Monday

Time: 10.30_{am} to 13.30_{p.m}

Maximum Marks: 70

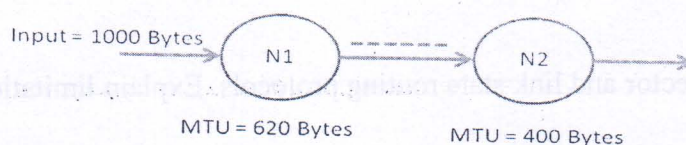
Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in the separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw the neat figures wherever required.

SECTION- I

Q.1

- A Why UDP is needed? Why can't a user program directly access IP? 2
- B Give the comparison of high speed networks with respect to following parameters: 4
- a. Data rate
 - b. Transmission Media
 - c. Access method
- C Consider the following topology of wired network: 6



How the original data bytes get fragmented. Show necessary calculation with TREE chart with respect to individual node.

Q.2

- A Draw the frame relay network to interface two networks with one has 802.3 and 802.5 topology. Discuss operation with routing tables. The frame relay network contains six switches to provide interface. Show the proper vales of DLCI 6
- B Draw a basic Delay loss system and derive basic equations of arrival rate and throughput using standard customer based model with respect to any network. 5
- OR
- A What is role of AAL? Explain AAL protocol and PDU generation along with affect of other layers 6
- B Draw the LAPF core formats for frame relay network an discuss explicit congestion problem in network 5

Q.3

- A** Draw IEEE 802.11 Protocol architecture and explain in brief with difference between PCF and DCF. 6
- B** Consider compressed video transmission in an ATM network. Suppose standard ATM cell must be transmitted through 5 switches. The data rate is 43Mbps. 6
- What is the transmission time for one cell through one switch?
 - If the switch is busy transmitting a cell, our cell has to wait until the other cell completes the transmission and if switch is free our cell is transmitted immediately. What is the maximum time from when a typical video cell arrives at the first switch until it is finished being transmitted by the fifth and last one.

OR

- A** What is IP addressing? Mentioned the limitations of existing IP addressing scheme with IPv4. Discuss different classes with example 6
- B** How do the encapsulations work in TCP/IP network? An application of TFTP is base used between client and server. The TFTP client transfer 400 bytes of data. The size of 4bytes of header gets added. It uses UDP protocol at transport layer. Calculate the total number of bytes which sent to server through physical layer. Also draw a TFTP client and server layer diagram. 6

SECTION – II

Q.4

- A** Compare distance vector and link state routing protocols. Explain limitations of each with example. 6
- B** Consider the graph with 6
- $V = \{a, b, c, d, e, f\}$ and
- $E = \{(a, b), (a, d), (b, c), (b, e), (c, e), (c, f), (d, e), (e, f)\}$
- Draw the Graph.
 - Find all the path from a to f
 - What is the length of the shortest path from a to f

Q.5

- A** Draw basic mechanism for congestion control with source and destination node. Explain following terms with respect to basic mechanism: 6
- Choke Packet
 - Implicit Congestion Signaling
 - Explicit Congestion Signaling
- B** Explain the operation of MPLS within a domain of MPLS enabled routers .Discuss key elements of operation with necessary diagram. 5

OR

- A Draw and explain RIP packet format and Its limitations 5
- B Draw RSVP host model and mentioned different events with protocol mechanism. Explain in brief. 6

- Q.6 Write short notes: (ANY TWO) 12
- i) Frame relay congestion control
 - ii) Gigabit Ethernet cabling standard
 - iii) RSA Algorithm
 - iv) Transperent and Non Transparent Fragmentation
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SECTION 1

Q.1 Why is CSMA/CD not used in wireless LANs? Why can't it be used practically directly across RF?

Q.2 Draw the representation of a networked network with respect to a binary parameter.

Q.3 Draw the network model.

Q.4 Draw a network.

Q.5 Consider the following topology of a network.



Q.6 How is original data bytes get fragmented. Show necessary calculation with TKB size with respect to individual ends.

Q.7 Draw the frame relay network to interconnect two networks with one end 20.1.1 and other 20.1.1 topology. Discuss operation with routing device. The frame relay network connects six switches to provide interface. Name the proper value of DLCI.

Q.8 Draw a Delay Based system and derive basic equations of arrival rate and throughput using standard queueing theory model with respect to any network.

OR

Q.9 What is AAL? Explain AAL protocol and PDU generation along with effect of error events.

Q.10 Draw the LAPP code format for frame relay network for security problem. Discuss congestion problem in network.